Obs	pelvic incidence	pelvic tilt	lumbar lordosis angle	sacral slope	pelvic radius	grade of spondylolisthesis	Class_att
1	63.0278175	22.55258597	39.60911701	40.47523153	98.67291675	-0.254399986	Abnormal
2	39.05695098	10.06099147	25.01537822	28.99595951	114.4054254	4.564258645	Abnormal
3	68.83202098	22.21848205	50.09219357	46.61353893	105.9851355	-3.530317314	Abnormal
4	69.29700807	24.65287791	44.31123813	44.64413017	101.8684951	11.21152344	Abnormal
5	49.71285934	9.652074879	28.317406	40.06078446	108.1687249	7.918500615	Abnormal
6	40.25019968	13.92190658	25.1249496	26.32829311	130.3278713	2.230651729	Abnormal
7	53.43292815	15.86433612	37.16593387	37.56859203	120.5675233	5.988550702	Abnormal
8	45.36675362	10.75561143	29.03834896	34.61114218	117.2700675	-10.67587083	Abnormal
9	43.79019026	13.5337531	42.69081398	30.25643716	125.0028927	13.28901817	Abnormal
10	36.68635286	5.010884121	41.9487509	31.67546874	84.24141517	0.664437117	Abnormal

The CONTENTS Procedure

Data Set Name	SAS.SPINE	**Observations**	310
Member Type	DATA	**Variables**	7
Engine	V9	**Indexes**	0
Created	10/27/2026 21:25:16	**Observation Length**	56
Last Modified	10/27/2026 21:25:16	**Deleted Observations**	0
Protection		**Compressed**	NO
Data Set Type		**Sorted**	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	2334
Obs in First Data Page	310
Number of Data Set Repairs	0
Filename	/home/u64368263/personal_projects/data/spine.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	15703732552
Access Permission	rw-r--r--
Owner Name	u64368263
File Size	256KB
File Size (bytes)	262144

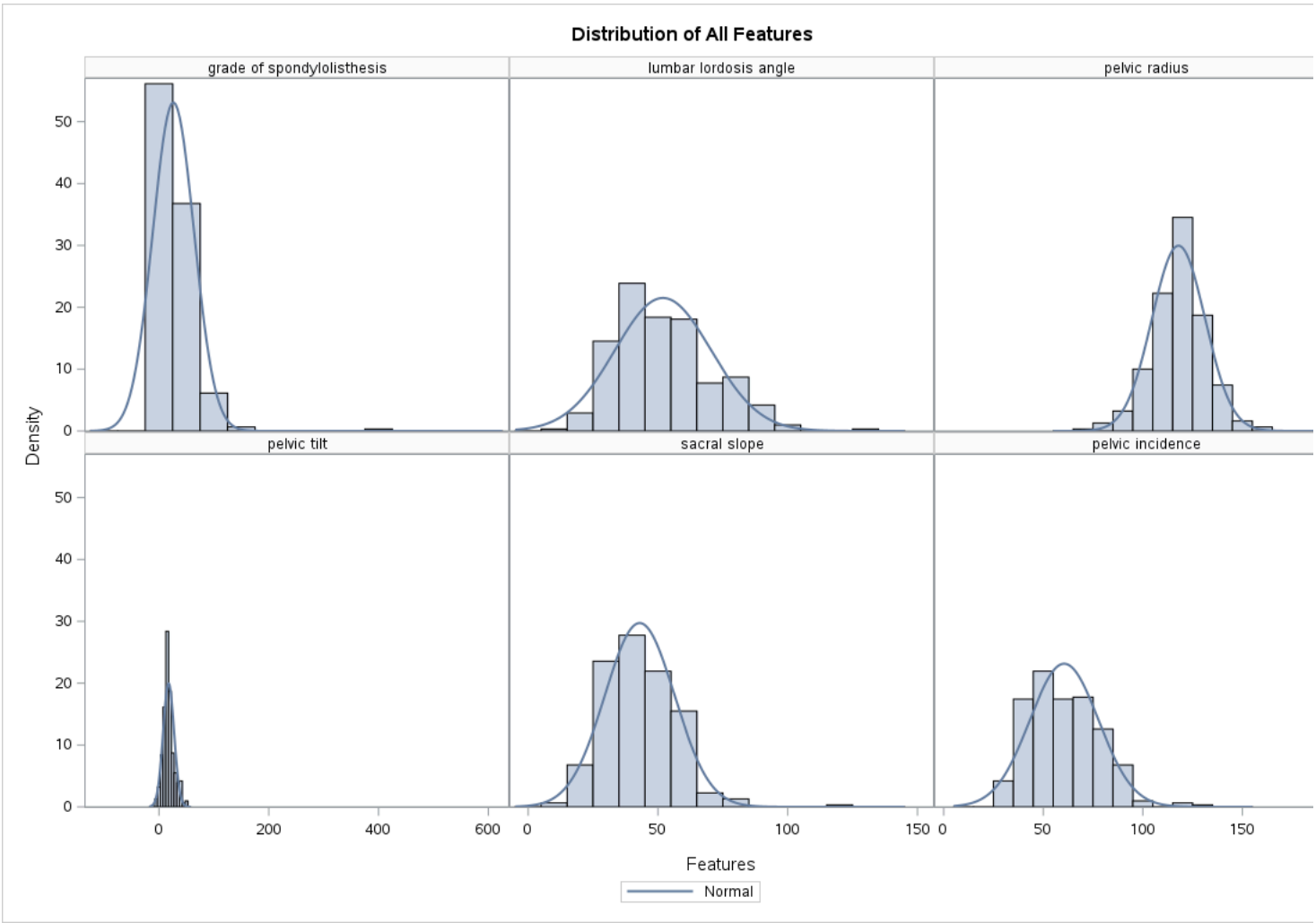
Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
6	grade of spondylolisthesis	Num	8	BEST12.	BEST32.
3	lumbar lordosis angle	Num	8	BEST12.	BEST32.
5	pelvic radius	Num	8	BEST12.	BEST32.
2	pelvic tilt	Num	8	BEST12.	BEST32.
4	sacral slope	Num	8	BEST12.	BEST32.
7	Class_att	Char	8	$8.	$8.
1	pelvic incidence	Num	8	BEST12.	BEST32.

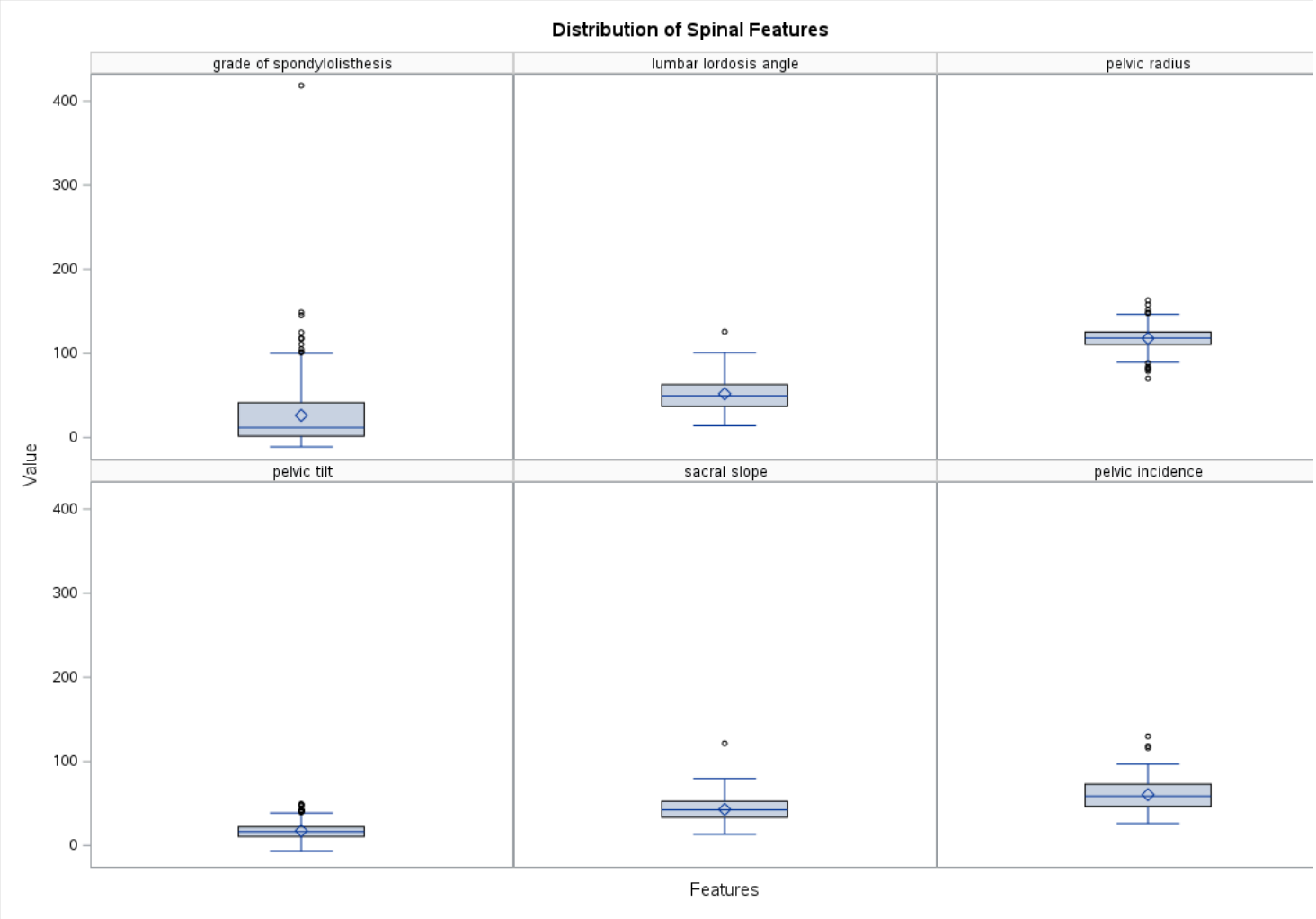
The MEANS Procedure

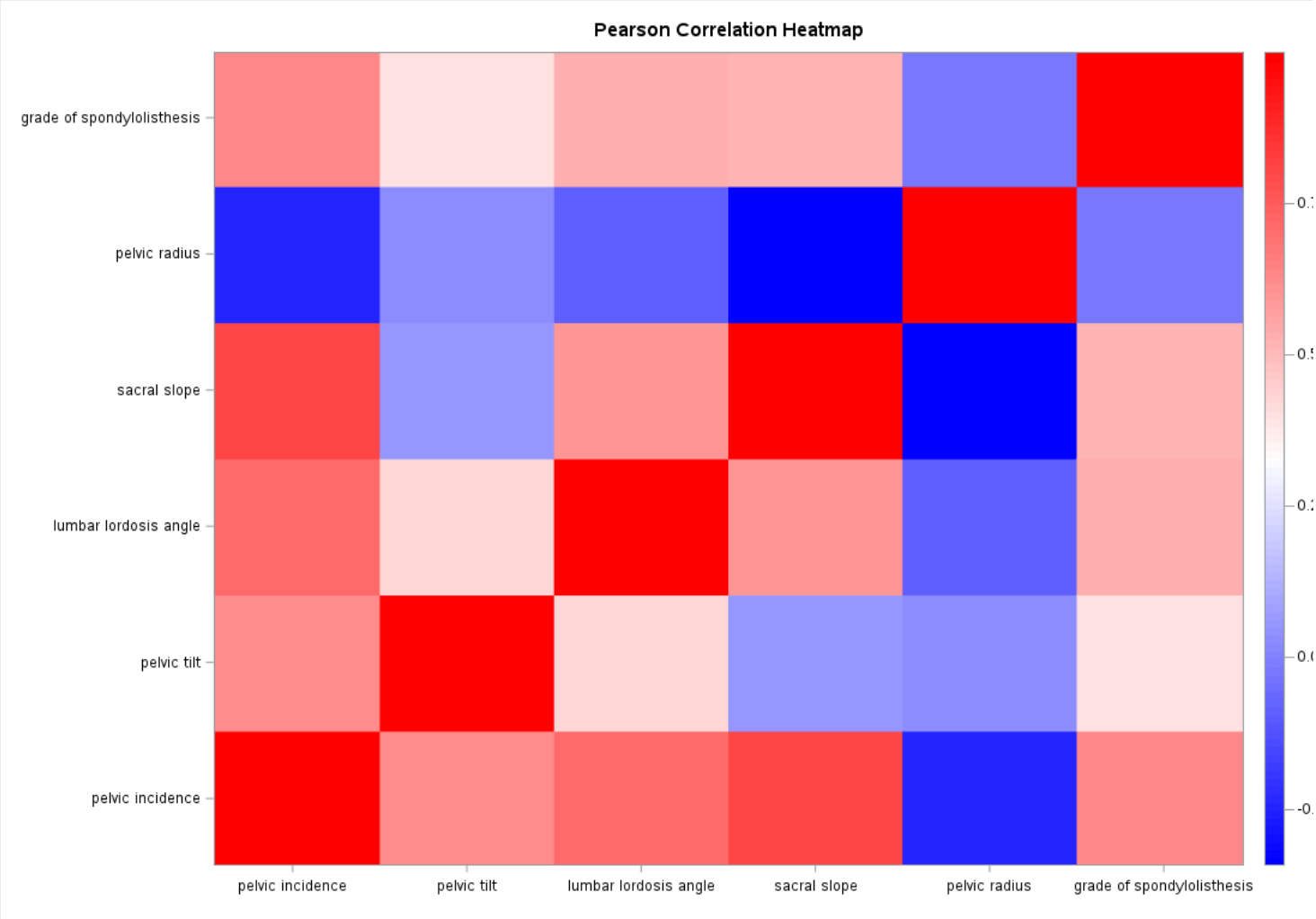
Variable	N	N Miss
pelvic incidence	310	0
pelvic tilt	310	0
lumbar lordosis angle	310	0
sacral slope	310	0
pelvic radius	310	0
grade of spondylolisthesis	310	0

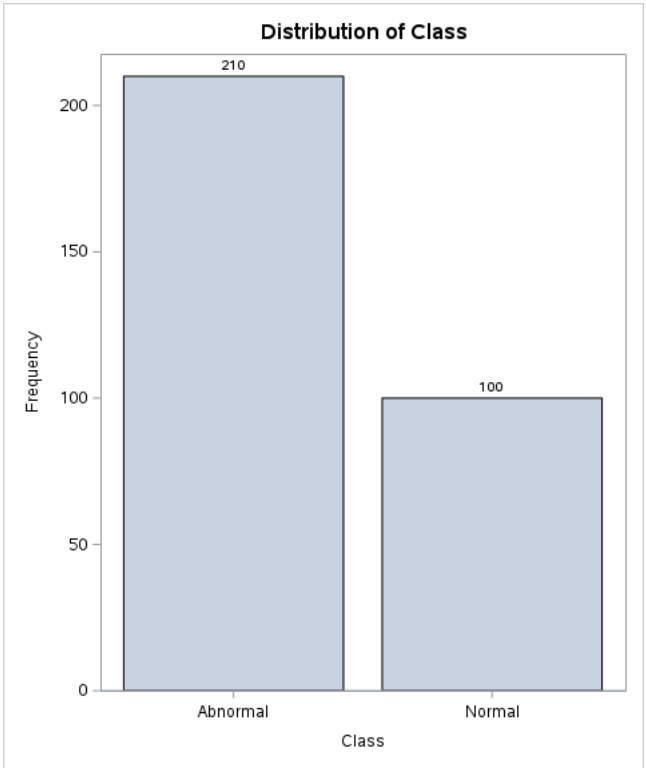
The FREQ Procedure

Class_att	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Abnormal	210	67.74	210	67.74
Normal	100	32.26	310	100.00


Distribution of All Features
grade of spondylolisthesis
lumbar lordosis angle
pelvic radius
pelvic tilt
sacral slope
pelvic incidence
Density
Features
Normal


Distribution of Spinal Features
grade of spondylolisthesis
lumbar lordosis angle
pelvic radius
pelvic tilt
sacral slope
pelvic incidence
Value
Features
0
100
200
300
400


Pearson Correlation Heatmap
grade of spondylolisthesis
pelvic radius
sacral slope
lumbar lordosis angle
pelvic tilt
pelvic incidence
pelvic incidence
pelvic tilt
lumbar lordosis angle
sacral slope
pelvic radius
grade of spondylolisthesis


Distribution of Class
210
100
200
150
100
50
0
Frequency
Abnormal
Normal
Class